Mathematical Analysis Notes

Shaojun Zan

6 July 2025

Contents

1 Sets and Numbers

1.1 Sets and Subsets

A **set** is a collection of objects, called **elements** or members.

If an element x is in the set S, we write $x \in S$. Otherwise, we write $x \notin S$ for x is not is S.

The number of elements in the set S is referred to as the **cardinality** or size, denoted as $|S|$.

If A and B are sets, then A is a subset of B if every x in A is also in B. That is, $x \in A \Rightarrow x \in B$, and we write

$$A \subseteq B, \quad \text{or equivalently} \quad B \supseteq A. \tag{1}$$

If there are elements in B that are not in A, we say A is a **proper subset** of B, denoted as

$$A \subset B, \quad \text{or equivalently} \quad B \supset A. \tag{2}$$

Written in mathematical language, we have

$$\begin{cases} A \subseteq B, \quad B \supseteq A: \quad x \in A \Rightarrow x \in B; \\ A \subset B, \quad B \supset A: \quad x \in A \Rightarrow x \in B, \text{ but } \exists y \in B \text{ such that } y \notin A. \end{cases} \tag{3}$$

There are several properties of this subset operation.

- $A \subseteq A$;
- $A \subseteq B$, $B \subseteq A \Rightarrow A = B$ — a good way to prove $A = B$;

  The equality is defined simply via **membership**. This implies that the sets $\{1, 2\}$, $\{2, 1\}$, $\{1, 2, 1, 2\}$ all are equal.
- $A \subseteq B$, $B \subseteq C \Rightarrow A \subseteq C$ — $\subseteq$ is transitive.

Clearly, $\subseteq$ is an **order relation** (order relations for numbers a, b: either $a \leq b$ or $b \leq a$). However, sometimes neither $A \subseteq B$ nor $B \subseteq A$ is true. For this reason, the relation $\subseteq$ is called a **partial ordering** of sets.

1.2 Operations

1.2.1 Union

$$A \cup B = \left\{ x \middle| x \in A \text{ or } x \in B \right\} \tag{4}$$

1.2.2 Intersection

$$A \cap B = \left\{ x \middle| x \in A \text{ and } x \in B \right\} \tag{5}$$

By Venn diagrams, it's simple to derive the following equalities. However, remember that Venn diagrams themselves do not suffice to prove these results.

$$|A \cup B| = |A| + |B| = |A \cap B| \tag{6}$$

$$|A \cup B \cup C| = |A| + |B| + |C| - |A \cap B| - |A \cap C| - |B \cap C| + |A \cap B \cap C| \tag{7}$$

The operations of unions and intersections are respectively commutative and associative:

$$A \cup B = B \cup A, \qquad A \cap B = B \cap A; \tag{8}$$

$$(A \cup B) \cup C = A \cup (B \cup C), \qquad (A \cap B) \cap C = A \cap (B \cap C). \tag{9}$$

When mixed together, we get the distributive laws:

$$A \cap (B \cup C) = (A \cap B) \cup (A \cap C); \tag{10}$$

$$A \cup (B \cap C) = (A \cup B) \cap (A \cup C). \tag{11}$$

1.2.3 Difference

$$A \setminus B = \left\{ x \middle| x \in A \text{ and } x \notin B \right\} \tag{12}$$

1.2.4 Complements

We denote the **universal set** by U, and the empty set, $\emptyset$. Thus, the complement of A, A', is defined as

$$A' = U \setminus A = \left\{ x \middle| x \in U \text{ and } x \notin A \right\}. \tag{13}$$

Now, we are prepared to meet De Morgan's rules:

$$\begin{aligned}(A \cup B)' &= A' \cap B';\\ (A \cap B)' &= A' \cup B'.\end{aligned} \tag{14}$$

1.2.5 Cartesian Product

$$A \times B = \left\{(a, b) \,\middle|\, a \in A,\ b \in B\right\} \tag{15}$$

$A \times B$ is the set of ordered pairs $(a.b)$. As an example, the 2D plane of real numbers is the Cartesian product of two copies of the set of real numbers, that is, $\mathbb{R}^2 = \mathbb{R} \times \mathbb{R}$.

1.3 Functions (Maps)

A function or map is a rule which, for every element of a set D (**domain**), uniquely assigns an element of another set T (**target** or **codomain**).

To specify a map f, we write

$$\begin{aligned} f : D &\to T \\ x &\to f(x), \end{aligned} \tag{16}$$

where $x \in D$, $f(x) \in T$.

There's another concept called **image** or **range**:

$$\text{Image}(f) = \left\{y \in T \middle| \exists x \in D \text{ such that } f(x) = y\right\}. \tag{17}$$

From this we would realize

$$\text{Image}(f) \subseteq T. \tag{18}$$

Importantly, there isn't a concept that corresponds to domain the same way image does to target (i.e. some subset of domain or domain as some subset).

Is $f(x)$ a map?

$f(x) = 1/x$ does not define a function $f : \mathbb{R} \to \mathbb{R}$, since $f(0)$ is undefined.

However, with the domain modified to $\mathbb{R} \setminus \{0\}$, $f(x) = 1/x$ is a map again.

1.3.1 Surjective / Onto

f is **surjective** iff $\forall y \in T$, $\exists x \in D$ such that $f(x) = y$ — no target element is missed out. $\Rightarrow$ Image$(f) = T$ for a surjective f.

1.3.2 Injective / One-to-one

f is **injective** if $x \neq y$ implies $f(x) \neq f(y)$, $\forall x, y \in D$. Equivalently, $f(x) = f(y)$ implies $x = y$ (contrapositive) — Every domain element is uniquely paired with one in the target.

1.3.3 Bijective

f is **bijective** if it's both surjective and injective.

For a bijective map, Every domain element is uniquely paired with one in the target, and vice versa.

Inverse maps exist for bijective maps:

$$\begin{aligned} f^{-1} : T &\to D \\ y &\to x(y). \end{aligned} \tag{19}$$

1.3.4 Theorem: Cardinality and Bijection

Let $f : A \to B$ be a map between finite sets A and B.

- f is a surjection: $|A| \geq |B|$;
- f is an injection: $|A| \leq |B|$;
- f is a bijection: $|A| = |B|$.

  This gives us another way to prove that two sets have the same cardinality: $|A| = |B|$ if $\exists f : A \to B$ such that f is a bijection.

1.3.5 The Pigeonhole Principle

If $f : A \to B$ for finite sets A and B where $|A| > |B|$, then f cannot be an injection.

1.3.6 Composition

Let $f : X \to Y$ and $g : Y \to Z$. Then, the composition of g and f is

$$\begin{aligned} g \circ f : X &\to Z \\ x &\to g \circ f(x) = g\left(f(x)\right). \end{aligned} \tag{20}$$

Functional composition wouldn't be so complicated if you keep track of the domain and target.

Functional composition is associative:

$$(h \circ g) \circ f = h \circ (g \circ f)\,. \tag{21}$$

2 Numbers

2.1 General Methods of Proof

2.1.1 Proof by Contradiction

We'll need the following truth table.

p	q	$p \to q$
T	T	T
T	F	F
F	T	T
F	F	T

The statement $p \to q$ means "if p, then q", and is only false when p is true but q is false.

Why Proof by Contradiction?

Let's consider proving the statement that $\sqrt{6} - \sqrt{2} > 1$. Without proof by contradiction, we have p as $\sqrt{6} - \sqrt{2} > 1$. Let's see what it takes us to:

$$\sqrt{6} - \sqrt{2} > 1 \quad \Rightarrow \quad 6 + 2 - 2\sqrt{12} > 1 \quad \Rightarrow \quad 4\sqrt{3} < 7 \quad \Rightarrow \quad 48 < 49\ (q).$$

This is to say, we have "if $\sqrt{6} - \sqrt{2} > 1$, then $48 < 49$" (this sentence is $p \to q$). Clearly, $p \to q$ is true, as we go from p and get q step by step.

Also, q, saying that $48 < 49$, is trivially true.

However, that both q and $p \to q$ are true does not give us information about the truth of p, i.e. if $\sqrt{6} - \sqrt{2} > 1$, as is written in the truth table.

Proof. The above explains why we need to use proof by contradiction. To begin, suppose $\sqrt{6} - \sqrt{2} \leq 1$ ($\neg p$).

The same logic leads us to $48 \geq 49$, which is trivially false. However, as we know $\neg p \to q$ is true, the truth table requires that $\neg p$ is false. Therefore, we have proved that $\sqrt{6} - \sqrt{2} > 1$. □

To examine if p is true, take $\neg p$, which leads to a q that is probably wrong. $\neg p \to q$ is true means that $\neg p$ is false, and thus p is true.

2.1.2 Proof by Induction

- $P(1)$ is true;
- For each $r \in \mathbb{N}$, if $P(r)$ is true, then $P(r+1)$ is true.

Alternatively, sometimes we need to prove with a step of 2:

- $P(1)$ and $P(2)$ are both true;
- For each $r \in \mathbb{N}$, if $P(r)$ is true, then $P(r+2)$ is true.

The idea is that we make sure every $P(k)$ is true, $\forall k \in \mathbb{N}$.

2.2 Countability and Uncountability

A set A is countable if there exists a bijection $f : \mathbb{N} \to A$.

While this definition may lead to some confusion, this bijection to natural numbers means that we may write this countable set in a list a_1, a_2, a_3, ..., where the indices are exactly the natural numbers.

Clearly, the natural number set itself is countable. Any infinite set $S \subset \mathbb{N}$ is also countable, like the set of primes. So, a bijection to a set $S \subset \mathbb{N}$ is also enough to prove countability.

2.3 Primes

2.3.1 Theorem: Infinite Number of Primes

There is an infinite number of primes.

Proof. Use proof by contradiction. Suppose there is a finite number of primes, and we may write them in a list: $\{p_1, p_2, \ldots, p_n\}$.

Now consider the number $p_{n+1} = p_1 p_2 \ldots p_n + 1$.

None of the listed primes divide p_{n+1}, as they all leave a remainder of 1. So, $p_n + 1$ is also a prime.

Therefore, we've reached a contradiction. □

2.3.2 Fundamental Theorem of Arithmetic

Every number has a unique factorization as products of prime numbers.

Proof. For a given N, divide repeatedly by p_1 until no longer possible. This fixes a. Then, repeat this process with p_2, which fixes b. So, we will finally get $N = p_1^a \times p_2^b \times \ldots$.

To see that this factorization is unique, suppose there was another factorization with powers a', b', c', ... Then, we must have

$$p_1^a \times p_2^b \times p_3^c \times \cdots = p_1^{a'} \times p_2^{b'} \times p_3^{c'} \times \ldots. \tag{22}$$

By simply canceling out powers of the primes on each side, it is readily seen that $a = a'$, $b = b'$, $c = c'$, etc. ☐

$n! + 1$ can be prime for some n, but $n! + 2$, $n! + 3$, ..., $n! + n$ are all composites.

By this theorem, every positive integer can be identified as

$$N = p_1^a \times p_2^b \times \ldots \quad \Rightarrow \quad \ln N = a \ln p_1 + b \ln p_2 + \ldots. \tag{23}$$

2.4 Rational and Irrational Numbers

We denote all the integers by $\mathbb{Z}$. The natural numbers are

$$\mathbb{N} = \left\{ n \middle| n \in \mathbb{Z},\ n > 0 \right\}. \tag{24}$$

The rational numbers are denoted by $\mathbb{Q}$:

$$\mathbb{Q} = \left\{ \frac{m}{n} \middle| m, n \in \mathbb{Z},\ n > 0 \right\}. \tag{25}$$

The rational numbers are countable.

Proof. Consider $\frac{m}{n}$ where m, $n \in \mathbb{N}$.

We can associate $\frac{m}{n}$ with the number $2^m 3^n$. By fundamental theorem of arithmetic, this gives a bijection to a subset of $\mathbb{N}$. ☐

There is a bijection from $\mathbb{N}$ to $\mathbb{N} \times \mathbb{N}$. In this logic, $\mathbb{N} \times \mathbb{N} \times \cdots \times \mathbb{N}$ is also countable.

Unions of an infinite number of infinite but countable sets are also countable.

2.4.1 Theorem: $\sqrt{2}$ is Irrational

Proof. Assume $\sqrt{2} = \frac{m}{n}$ is rational, where not both m and n are even.

This gives us $2 = \frac{m^2}{n^2} \Rightarrow m^2 = 2n^2$. Therefore, m^2 is even and so is m, and $m = 2k$ for $k \in \mathbb{Z}$.

Therefore, $4k^2 = 2n^2$ gives that n is also even. We've reached a contradiction. ☐

$\sqrt{3}$ is also irrational.

Proof. Suppose $\sqrt{3}$ is rational with $\sqrt{3} = \dfrac{p}{q}$, where p, q are in lowest terms.

So, $3 = \dfrac{p^2}{q^2}$, $p^2 = 3q^2 \Rightarrow p^2$ is a multiple of 3.

- Let $p = 3k + 1$ where $k \in \mathbb{Z}$.

  Clearly, $p^2 = 9k^2 + 6k + 1 = 3\left(3k^2 + 2k\right) + 1$ is not divisible by 3.

- Let $p = 3k + 2$ where $k \in \mathbb{Z}$.

  Clearly, $p^2 = 9k^2 + 12k + 4 = 3\left(3k^2 + 4k + 1\right) + 1$ is not divisible by 3.

Hence, the only possibility is $p = 3k$, where $k \in \mathbb{Z}$. Then, $9k^2 = 3q^2$, $q^2 = 3k^2$. Similarly, we have $q = 3r$ for some $r \in \mathbb{Z}$. This shows that p and q are not in their lowest terms. □

We may also use the fundamental theorem of arithmetic in our proof. For example, let's prove that $\sqrt{5}$ is irrational.

Proof. Suppose $\sqrt{5} = \dfrac{m}{n}$ is rational, where $m, n \in \mathbb{Z}$. This gives us that $m^2 = 5n^2$.

By fundamental theorem of arithmetic, $m = 2^a 3^b 5^c 7^d \ldots$, $n = 2^{a'} 3^{b'} 5^{c'} 7^{d'} \ldots$, where the powers are integers.

So, $m^2 = 5n^2$ means that $2^{2a} 3^{2b} 5^{2c} 7^{2d} \cdots = 2^{2a'} 3^{2b'} 5^{2c'+1} 7^{2d'} \ldots$.

Equating all the powers, we get $2c = 2c' + 1$. This makes a contradiction. □

2.4.2 Algebraic Numbers

$$A = \bigcup_{n=0}^{\infty} A_n, \quad A_n = \left\{x \middle| x \in \mathbb{R},\ p_n(x) = 0\right\}, \quad p_n(x) = \sum_{k=0}^{n} a_k x^k, \quad a_k \in \mathbb{Z} \tag{26}$$

In this definition, $\sqrt{2}$ is an algebraic number, as it satisfies $x^2 - 2 = 0$.

The algebraic numbers are countable.

Proof. The set of algebraic numbers is the union of all A_n. Therefore, A is countable if all A_n are countable.

A_n is specified by the coefficients $a_k \in \mathbb{Z}$, and the polynomials have solutions x_r, $r \in \mathbb{N}$. This implies that A_n is a subset of $\mathbb{Z}^n \times \mathbb{N}$ and is thus countable. So, the algebraic numbers are also countable. □

2.5 The Decimal Representation

$$x = a_0 + \frac{a_1}{10} + \frac{a_2}{100} + \cdots = a_0 + \sum_{n=1}^{\infty} \frac{a_n}{10^n} \tag{27}$$

2.5.1 Non-Uniqueness

As we've already known but never explained in this way, the fact that $0.999\cdots = 1.000\ldots$ is due to the fact that the decimal expansion is not unique.

2.5.2 Rational Numbers

The rational numbers are always represented by periodic decimals. Conversely, a periodic decimal is always a rational number.

As an example, consider $x = 0.\overline{b_1 b_2 \ldots b_k}$. Let $B = 0.b_1 b_2 \ldots b_k$.

So, $10^k x = 10^k B + x$, and this gives that

$$x = \frac{10^k B}{10^k - 1}. \tag{28}$$

2.5.3 Binary Representation

$$x = \sum_{n=1}^{\infty} \frac{a_n}{2^n} \quad \text{for } x \leq 1 \tag{29}$$

Binary representation isn't unique either: $(0.111\ldots)_2 = 1$.

2.5.4 Ternary Representation

$$x = \sum_{n=1}^{\infty} \frac{a_n}{3^n} \quad \text{for } x \leq 1 \tag{30}$$

Similarly, $(0.222\ldots)_3 = 1$.

2.5.5 Notes

The apparent use of this sum of rationals does not mean that these representations are a subset of the rationals $\mathbb{Q}$. They all apply to the reals $\mathbb{R}$.

2.6 Rational Approximations

Theorem: $\forall x \in \mathbb{R}$, $\exists \frac{p}{q} \in \mathbb{Q}$ such that $\left| x - \frac{p}{q} \right| \leq \frac{1}{q^2}$.

2.7 The Real Numbers

2.7.1 Uncountability

The real numbers are uncountable.

Proof. There are several ways to prove this fact.

- Suppose that the reals are countable. So, we can make a list of them:

$$r_1 = m_1.a_{11}a_{12}a_{13}\dots$$
$$r_2 = m_2.a_{21}a_{22}a_{23}\dots$$
$$r_3 = \dots.$$

  Now construct a number $0.b_1b_2b_3\dots$ by choosing $b_i = 1$ or 2 such that $b_i \neq a_{ii}$. Clearly, this new number is different from every number in the list. Therefore, a contradiction is reached.

- Consider the unit interval $0 \leq x \leq 1$.

  Clearly, there is a bijection that maps $\mathbb{R}$ to $[0,1]$. For example, $f(x) = \dfrac{\tanh x + 1}{2}$.

  Suppose that the reals are countable. So, we can make a list of them: $r_1, r_2, r_3, \dots$. Then, we surround each r_n with an interval of length 10^{-n}. So, with all the reals in the list, we've taken up an interval of length

$$\frac{1}{10} + \frac{1}{100} + \dots = \sum_{n=1}^{\infty} 10^{-n} = \frac{1}{9}.$$

  This is a contradiction, since we have, by hypothesis, included every point in the unit interval and surrounded each by a nonzero size.

- Consider the interval $[0,1]$ and suppose for contradiction that all points in the interval may be put in an ordered list $\{x_1, x_2, \dots\}$.

  Choose nested set of intervals $[0,1] \supseteq I_1 \supseteq I_2 \supseteq \dots$, where the intervals are chosen so that $x_1 \notin I_1$, $x_2 \notin I_2$, etc. such that every point has an interval which does not contain it.

  But, by the nested interval property, $\exists \xi \in [0,1]$ such that $\xi \in I_n$ for all n. There exists a number that sits in every interval, hence cannot belong in the original list.

  Hence, this is a contradiction.

□

2.8 Upper Bound

2.8.1 Definition

$\forall a \in A$, $\exists b \in \mathbb{R}$ such that $a \leq b \Rightarrow b$ is an upper bound for set A.

The upper bound does not have to be in the set.

2.8.2 Supremum / Least Upper Bound (LUB)

The LUB is b such that $b \leq c$ for any upper bound c. The LUB is unique.

Equivalently, we may define LUB via $\forall \epsilon > 0$, $\exists a \in A$ such that $b - \epsilon < a \leq b$, if b is the LUB.

2.9 Lower Bound

2.9.1 Definition

$\forall a \in A$, $\exists b \in \mathbb{R}$ such that $a \geq b \Rightarrow b$ is an lower bound for set A.

2.9.2 Infinum / Greatest Lower Bound (GLB)

The GLB is b such that $b \geq c$ for any lower bound c.

Equivalently, if b is the GLB, then $\forall \epsilon > 0$, $\exists a \in A$ such that $b \leq a < b + \epsilon$.

2.10 Completeness Axiom: "Completeness of the Reals"

If $A \subset \mathbb{R}$ is bounded from above, then A has a LUB in $\mathbb{R}$.

This axiom is not satisfied by the rationals.

2.11 Nested Set of Closed Intervals

$$I_n = [a_n, b_n] = \left\{ x \middle| a_n \leq x \leq b_n,\ x \in \mathbb{R},\ n \in \mathbb{N} \right\}, \tag{31}$$

where the endpoints are such that,

$$I_1 \supseteq I_2 \supseteq \cdots \supseteq I_n \supseteq I_{n+1} \supseteq \ldots. \tag{32}$$

2.12 Theorem: Nested Intervals

- For any nested set of closed intervals, $\exists \xi \in \mathbb{R}$ such that $\xi \in I_n$ for all n.
- If the intervals are such that $|a_n - b_n| \to 0$ as $n \to \infty$, then ξ is unique.

2.13 Theorem: Density of the Rationals and Irrationals

Between any two reals a and b there are both rational and irrational numbers.

Proof. Suppose $b > a$. We can find $n \in \mathbb{Z}$ such that $n > \dfrac{1}{b-a}$, which implies that $\dfrac{1}{n} < b - a$.

This implies that $\exists m \in \mathbb{Z}$ such that $a < \dfrac{m}{n} < b$.

Also, we know $\dfrac{1}{n\sqrt{2}} < \frac{1}{n} < b - a$. So, $\exists m \in \mathbb{Z}$ such that $a < \dfrac{m}{n\sqrt{2}} < b$. $\square$

2.14 The Cantor Set

The Cantor set is a set of nested subsets. Consider the initial interval $\Delta_0 = [0, 1]$. Remove the middle third, we get

$$\Delta_1 = \left[0, \frac{1}{3}\right] \bigcup \left[\frac{2}{3}, 1\right].$$

Remove the middle third of the intervals again, we get

$$\Delta_2 = \left[0, \frac{1}{9}\right] \bigcup \left[\frac{2}{9}, \frac{1}{3}\right] \bigcup \left[\frac{2}{3}, \frac{7}{9}\right] \bigcup \left[\frac{8}{9}, 1\right].$$

We can easily see that $\Delta_0 \supset \Delta_1 \supset \Delta_2 \supset \dots$.

Then, the Cantor set is defined as

$$\Delta = \bigcap_{k=1}^{\infty} \Delta_k, \tag{33}$$

which is **uncountable with zero geometric size**.

2.14.1 Geometric Perspective

The total length of interval we have removed is

$$L = \frac{1}{3} + \frac{2}{9} + \frac{4}{27} + \cdots = \frac{1}{3} \sum_{n=0}^{\infty} \left(\frac{2}{3}\right)^n = 1.$$

Therefore, the Cantor set has zero geometric size.

By ternary representation, $x = \sum_{n=1}^{\infty} \dfrac{a_n}{3^n}$ for $a_n = 0,\ 1,\ 2$. So, removing the middle third is equivalent to removing 1. The Cantor set involves all numbers whose ternary representation involves only 0 and 2.

However, by the non-uniqueness of ternary representation, we know that e.g. $(0.20222\dots)_3 = (0.21)_3$. 1 can appear as the last digit in Cantor ternary numbers.

2.14.2 Mathematical Perspective

The Cantor set involves strings of just 2 numbers, so it is equivalent to the binary representation, which has a bijection to $\mathbb{R}$. This is why the Cantor set is uncountable.

2.15 Power Sets of Infinite Sets

$$P(X) \equiv 2^X = \{A \big| A \subseteq X\} \tag{34}$$

Clearly,

$$|P(A)| = 2^{|A|} > |A| \quad \text{for finite } |A|. \tag{35}$$

By the theorem of bijection and cardinality, we know that there cannot exist a bijection from A to its power set $P(A)$. Therefore, the fact that A is countable tells us nothing about the countability of its power set.

There is no bijection between a non-empty set A and its power set, 2^A.

Proof.

- We may prove $|A| \leq |2^A|$ first.

  We can find $f : A \to 2^A$, where $f(a) = \{a\}$.

  Clearly, this is an injection, so $|A| \leq |2^A|$.

- Rule out $|A| = |2^A|$ by proof by contradiction.

  Suppose $\exists g : A \to 2^A$ is a bijection.

  Consider the set $R = \{a \in A \big| a \notin g(a)\}$.

  R is a subset of A, so $R \in 2^A$.

  So, $\exists y \in A$ such that $g(y) = R$.

  If $y \in R$, then $y \notin g(y) = R$. Contradiction.

  If $y \notin R$, then $y \in A$ but $y \notin g(y) = R \Rightarrow y \in R$. Contradiction.

□

One may label the elements of $P(A)$ using binary strings involving yes (Y) or no (N). Then, elements of $P(A)$ correspond to the 2^n binary strings of length n consisting of Y and N.

Then, for infinite set A, $P(A)$ bijects all infinite binary strings, which bijects the reals $\mathbb{R}$. Therefore, $|P(A)| = |\mathbb{R}|$.

Additionally, the power set of $\mathbb{R}$ is more uncountable than $\mathbb{R}$ itself, $|P(\mathbb{R})| > |\mathbb{R}|$.

2.16 Miscellaneous: e is Irrational

Proof. Suppose $\mathrm{e} = \dfrac{m}{n}$ is rational, where $m,\, n \in \mathbb{Z}$.

$$\mathrm{e}^x = \sum_{k=0}^{\infty} \frac{x^k}{k!} \quad \Rightarrow \quad \mathrm{e} = \sum_{k=0}^{\infty} \frac{1}{k!}.$$

Therefore, $m = n\mathrm{e} \Rightarrow m\,(n-1)! = \mathrm{e}n!$. Expanding the factorial,

$$m\,(n-1)! = n! \left[\left(2 + \frac{1}{2!} + \frac{1}{3!} + \cdots + \frac{1}{n!} \right) + \frac{1}{(n+1)!} + \ldots \right].$$

LHS is clearly an integer, while RHS is an integer plus a remainder R, where

$$R = \frac{1}{n+1} + \frac{1}{(n+1)(n+2)} + \ldots.$$

For $n \in \mathbb{N}$, $n \geq 1$, so

$$R \leq \frac{1}{2} + \frac{1}{6} + \cdots < \frac{1}{2} + \frac{1}{2^2} + \cdots = 1.$$

Therefore, $R < 1$, and this is a contradiction. $\square$

2.17 Classification of Real Numbers

$$\text{Real Numbers } (\mathbb{R}) \begin{cases} \text{Rational Numbers } (\mathbb{Q}) \\ \text{Irrational Numbers } (\mathbb{R} \setminus \mathbb{Q}) \end{cases} \tag{36}$$

In rational numbers, we can identify natural numbers $\mathbb{N}$, which may be classified into primes and composites. There are also integers $\mathbb{Z}$, etc.

$$\text{Real Numbers } (\mathbb{R}) \begin{cases} \text{Algebraic Numbers} \supset \mathbb{Q} \\ \text{Transcendental Numbers} \supset \mathbb{R} \setminus \mathbb{Q} \end{cases} \tag{37}$$

3 Sequences

3.1 Introduction

3.1.1 Definition

A sequence is a map from $\mathbb{N}$ into another set of numbers.

3.1.2 Convergent Sequence

$$|a_n - a| < \epsilon, \ \forall \epsilon > 0 \text{ if } n > N \quad \Rightarrow \quad \lim_{n\to\infty} a_n = a \tag{38}$$

Our important job here is to find $N(\epsilon)$.

The limit of a sequence is unique.

Proof. Consider $a_n \to a$ and $a_n \to b$ as $n \to \infty$. By the triangle inequality,

$$|a - b| = |a - a_n - b + a_n| \leq |a_n - a| + |b - a_n|.$$

RHS can be made arbitrarily small, so $a = b$. □

3.1.3 The Squeezing Theorem

$a_n \leq b_n \leq c_n$, with $a_n \to a$, $c_n \to a$, then b_n must converge to a.

Proof. Consider $c_n \leq b_n \leq a_n$ where $c_n \to a$ and $a_n \to a$.

Define $\bar{a}_n = a_n - a$, $\bar{b}_n = b_n - a$, $\bar{c}_n = c_n - a$. So,

$$|a_n - a| = \bar{a_n} < \epsilon, \ \forall \epsilon > 0 \text{ if } n > N, \text{ and the same for } c_n. \ (-|\bar{c}_n| > -\epsilon)$$

Therefore, $-|\bar{c}_n| \leq \bar{c}_n \leq \bar{b}_n \leq \bar{a}_n \leq |\bar{a}_n|$, $-\epsilon < \bar{b}_n < \epsilon \Rightarrow \left|\bar{b}_n\right| < \epsilon$.

So, $|b_n - a| < \epsilon$, $\forall \epsilon > 0$ if $n > N$. b_n converges to a. □

3.1.4 Theorem: Ratio Test for Sequences

$$L = \lim_{n\to\infty} \left| \frac{x_{n+1}}{x_n} \right| \tag{39}$$

(Written in epsilon language, $\forall \epsilon > 0$, $\left| \left| \frac{x_{n+1}}{x_n} \right| - L \right| < \epsilon$ if $n > N$.) If $L < 1$, then the sequence converges absolutely.

Proof. By definition of L, $\forall\epsilon > 0$, $\exists N$ such that $\left|\dfrac{x_{n+1}}{x_n} - L\right| < \epsilon$.

This gives $L - \epsilon < \dfrac{x_{n+1}}{x_n} < L + \epsilon$.

Consider $L < 1$. $\exists\epsilon > 0$ such that $R = L + \epsilon < 1$. So, $0 < \dfrac{x_{n+1}}{x_n} < R < 1$, $x_{n+1} < Rx_n$.

Therefore, $0 < x_{n+1} < Rx_n < \cdots < R^n x_1$.

As we know $R^n \to 0$ as $n \to \infty$, $x_{n+1} \to 0$ by the squeezing theorem.

For $L > 1$, $\exists\epsilon > 0$ such that $L - \epsilon > 1$.

To show $L = 1$ is indecisive, consider $x_n = n$ and $x_n = 1/n$. □

3.2 Monotone Sequences

3.2.1 Theorem

A convergent sequence is bounded.

Proof. For a convergent sequence, $|a_n - a| < \epsilon$, $\forall\epsilon > 0$ if $n > N$.

This gives $|a_n| = |(a_n - a) + a| \leq |a_n - a| + |a| < \epsilon + |a|$. So, this sequence is bounded for $n > N$ with any $\epsilon > 0$. For $n \leq N$,

$$|a_n| \leq \max\{|a_1|, |a_2|, \ldots, |a_n|\}.$$

Therefore, $|a_n|$ is bounded for all n. □

3.2.2 Theorem: Bounded Monotone Sequences — Proof by Completeness Axiom

If a sequence is monotonically increasing and bounded above, then its must be convergent.

Proof. Use completeness axiom. Suppose $\mathrm{LUB}(\{a_n\}) = a$.

By the definition of LUB, $\forall\epsilon > 0$, $\exists N \in \mathbb{N}$ such that $a - \epsilon < a_N \leq a$.

As the sequence is monotonically increasing, $a_n \geq a_N$ for $n > N$. So, $a - \epsilon < a_n$.

Noting that $a_n \leq a$, $|a_n - a| < \epsilon$, $\forall\epsilon > 0$ if $n > N$. Therefore, a_n converges to a. □

3.2.3 Theorem — Proof by Bounded Monotone Sequences

Consider nested intervals in $\mathbb{R}$ defined by $I_n = [a_n, b_n]$.

$\exists\xi \in \mathbb{R}$ such that $\xi \in I_n$ for all n.

If $|a_n - b_n| \to 0$ as $n \to \infty$, then ξ is unique.

Proof. The lower end of each interval defines a monotonically increasing sequence, $a_n \leq a_{n+1}$. The upper end of each interval defines a monotonically decreasing sequence, $b_{n+1} \leq n_n$.

As $a_n \leq b_n \leq b_1$, $a_1 \leq a_n \leq b_n$, we see that a_n is bounded from above, and b_n is bounded from below.

Hence, $a_n \to a$, $b_n \to b$. We thus have $a \leq b$, and $\exists \xi \in [a, b]$, $\xi \in I_n$ for all n.

To prove uniqueness, use $|a_n - b_n| \to 0$ to show $b_n \to a$:

$$|b_n - a| = |b_n - a_n + a_n - a| \leq |b_n - a_n| + |a_n - a| < \epsilon_1 + \epsilon_2,\ \forall \epsilon_1,\ \epsilon_2 > 0 \text{ if } n > N.$$

Therefore, $b_n \to a$, we have $\xi = a$ uniquely. $\square$

3.3 Subsequences and Cauchy Series

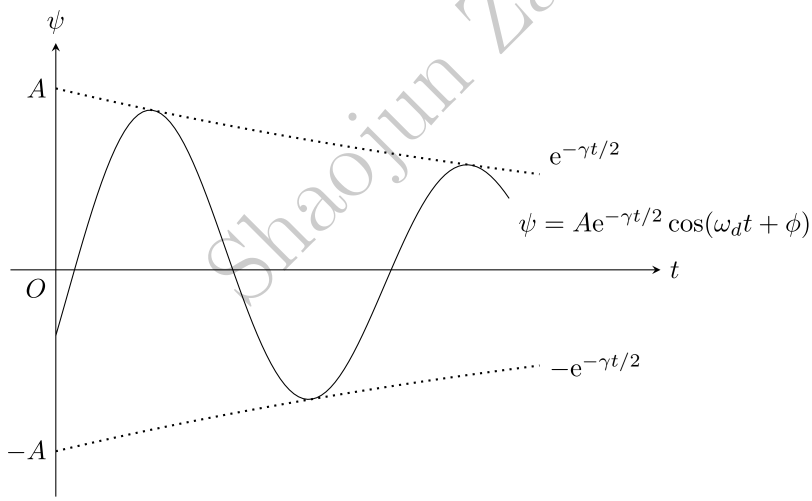
3.3.1 Definition

Let $\{a_n\}$ be a sequence and let n_i, $n_1 < n_2 < n_3$ be a strictly increasing sequence in $\mathbb{N}$.

Then, $\{b_n\}$ defined by $b_i = a_{n_i}$ is called a **subsequence** of $\{a_n\}$.

3.3.2 Theorem of Subsequences

Every sequence has a subsequence that is monotonically increasing or decreasing.



To see a monotonically decreasing subsequence, simply choose the peaks of this damped oscillator.

3.3.3 Theorem: Bolzano-Weierstrass

Every bounded sequence has a convergent subsequence in $\mathbb{R}$.

To see this, we may find a monotonous subsequence. As this sequence itself is bounded, the subsequence should also be bounded. That this subsequence is bounded and monotonous means that it is convergent.

3.3.4 Cauchy Sequence Definition

$$|a_n - a_m| < \epsilon,\ \forall \epsilon > 0 \text{ if } n, m > N. \tag{40}$$

Note that there is no need to find a such that $\lim_{n\to\infty} a_n = a$.

3.3.5 Theorem: Cauchy Sequence

Every Cauchy sequence is bounded.

Proof. By definition, $|a_n - a_m| < \epsilon$, $\forall \epsilon > 0$ if n, $m > N$.

Take $m = N$, $\epsilon = 1$, then $|a_n - a_N| < 1$. So,

$$|a_n| = |a_n - a_N + a_N| \le |a_n - a_N| + |a_N| < 1 + |a_N| \text{ for } n \ge N.$$

For $n \le N$, $|a_n| \le \max\{|a_1|, |a_2|, \ldots, |a_N|\}$.

Therefore, $|a_n|$ is bounded for all n. □

3.3.6 Theorem

$$\text{Convergence} \Leftrightarrow \text{Cauchy Sequence} \tag{41}$$

3.4 Contractive Sequences

3.4.1 Definition

$$\forall n \in \mathbb{N},\ \exists K \in (0,1) \text{ such that } |x_{n+1} - x_n| \le K|x_n - x_{n-1}| \tag{42}$$

3.4.2 Theorem

A contractive sequence is a Cauchy sequence (and converges in $\mathbb{R}$).

Proof. By definition,

$$|x_{n+1} - x_n| \le K|x_n - x_{n-1}| \le K^2|x_{n-1} - x_{n-2}| \le \cdots \le K^{n-1}|x_2 - x_1|.$$

Thus,

$$\begin{aligned}
|x_m - x_n| &= |x_m - x_{m-1} + x_{m-1} - \cdots - x_{n+1} + x_n| \\
&\le |x_m - x_{m-1}| + |x_{m-1} - x_{m-2}| + \cdots + |x_{n+1} - x_n| \\
&\le \left(K^{m-2} + K^{m-3} + \cdots + K^{n-1}\right)(x_2 - x_1) \\
&= K^{n-1}\left(1 + K + \cdots + K^{m-n-1}\right)|x_2 - x_1| \\
&= K^{n-1}\frac{1 - K^{m-n}}{1 - K}|x_2 - x_1|.
\end{aligned}$$

This clearly goes to 0 as $m > n$ as $n \to \infty$. So, the sequence is Cauchy.
Alternatively, $K^{n-1}\left(1 + K + \cdots + K^{m-n-1}\right) < K^{n-1}\sum_{j=0}^{\infty} K^j = \frac{K^{n-1}}{1-K}$.
To show $\frac{K^{n-1}}{1-K}|x_2 - x_1| < \epsilon$ for any $\epsilon > 0$, find $N(\epsilon)$. Therefore,

$$\begin{aligned}
K^{n-1} &< \frac{\epsilon(1-K)}{|x_2 - x_1|} \\
(n-1)\ln K &> \ln\frac{\epsilon(1-K)}{|x_2 - x_1|} \text{ (sign change because } 0 < K < 1) \\
n &> \ln\frac{\epsilon(1-K)}{|x_2 - x_1|} / \ln K + 1 = \log_K \frac{\epsilon(1-K)}{|x_2 - x_1|} + 1.
\end{aligned}$$

□

3.5 Summary

$$\text{Sequences } (f : \mathbb{N} \to A) \begin{cases} \text{divergent} \\ \text{convergent} \ \Leftrightarrow \ \text{Cauchy sequence} \begin{cases} \text{Contractive sequence} \\ \cdots \end{cases} \end{cases} \tag{43}$$

$$\begin{cases}
\text{Convergence} & \forall\epsilon > 0,\ |a_n - a| < \epsilon \text{ if } n > N \\
\text{Cauchy sequence} & \forall\epsilon > 0,\ |a_n - a_m| < \epsilon \text{ if } n, m > N \\
\text{Contractive sequence} & \forall n \in \mathbb{N},\ \exists K \in (0,1) \ \text{ such that } |x_{n+1} - x_n| \le K|x_n - x_{n-1}|
\end{cases} \tag{44}$$

If convergence is hard to prove (i.e. a is hard to find), consider proof via a Cauchy sequence, which does not require knowledge of a.

4 Series

4.1 Definition

4.1.1 From Partial Sum to Series

$$s_n = \sum_{k=1}^{n} a_k \quad \Rightarrow \quad s = \lim_{n\to\infty} s_n \tag{45}$$

4.1.2 Convergence

$$\left| s - \sum_{k=1}^{n} a_k \right| < \epsilon,\ \forall \epsilon > 0 \text{ if } n > N \tag{46}$$

4.1.3 Cauchy Series

Convergence $\Leftrightarrow$ Cauchy series.

$$\left| \sum_{k=n}^{m} a_k \right| < \epsilon,\ \forall \epsilon > 0 \text{ if } n, m > N \tag{47}$$

4.1.4 Absolute Convergence

By absolute convergence, we mean that $\sum_{k=1}^{\infty} |a_k|$ converges.

4.2 Tests for Convergence

4.2.1 Theorem (Proof by Cauchy Series)

$$\sum_{k=1}^{\infty} a_k \text{ converges} \quad \Rightarrow \quad \lim_{k\to\infty} a_k = 0. \tag{48}$$

Proof. The series is convergent, so is a Cauchy series. By definition,

$$\left| \sum_{k=n}^{m} a_k \right| < \epsilon,\ \forall \epsilon > 0 \text{ if } n, m > N.$$

Now take $m = n$. So, $|a_n| < \epsilon$ for $n > N$. Hence, $a_n \to 0$ as $n \to \infty$. $\square$

4.2.2 Theorem: The Comparison Test

Let $\sum_{k=1}^{\infty} c_k$ be a convergent series of positive terms. Then, any other series $\sum_{k=1}^{\infty} a_k$ converges absolutely if $|a_k| < c_k$.

Proof. The series $\sum_{k=1}^{\infty} c_k$ is convergent, so it is a Cauchy series. Therefore,

$$\sum_{k=n}^{m} |a_k| < \sum_{k=n}^{m} c_k = \sum_{k=n}^{m} |c_k| < \epsilon,\ \forall \epsilon > 0 \text{ if } n, m > N.$$

Thus, $\sum_{k=n}^{m} |a_k|$ is a Cauchy series, and it converges. □

4.2.3 Theorem: Ratio Test

$$\rho = \lim_{n\to\infty} \left| \frac{a_{n+1}}{a_n} \right| \tag{49}$$

If $\rho < 1$, then the series converges. If $\rho > 1$, then the series diverges. $\rho = 1$ is indecisive.

4.2.4 Theorem: Root Test

$$\rho = \lim_{n\to\infty} |a_n|^{1/n} \tag{50}$$

If $\rho < 1$, then the series converges. If $\rho > 1$, then the series diverges. $\rho = 1$ is indecisive.

4.3 Conditionally Convergent Series

4.3.1 Theorem: Alternating Series Test

Let a_n be a monotonically decreasing sequence of positive numbers with $a_n \to 0$ as $n \to \infty$. Then, the series $\sum_{k=1}^{\infty} (-1)^{k+1} a_k$ converges.

4.3.2 Theorem: Dirichlet's Test

If the sequence $\{a_n\}$ is monotonically decreasing with $a_n \to 0$ as $n \to \infty$, and if the partial sums $t_n = \sum_{k=1}^{n} b_k$ are bounded, $|t_n| \leq B$, for some B and for all n, then $\sum_{n}^{\infty} a_n b_n$ converges.

4.4 Riemann's Reordering Theorem

The sum of a conditionally convergent series depends on the order of the terms. It can converge to any real number.

4.5 Power Series

4.5.1 Definition

A power series is a series with variables (i.e. x):

$$\sum_{n=0}^{\infty} a_n x^n. \tag{51}$$

The convergence of a power series depends on x.

4.5.2 Theorem: Radius of Convergence

There exists a real number R, the radius of convergence, such that $|x| < R$ implies absolute convergence and $|x| > R$ implies divergence. There are no general results for the case $x = \pm R$ and these require special investigation.

4.6 Series and Integrals

4.6.1 Theorem: Integral Comparison Test

Consider $\displaystyle\int_1^{\infty} \mathrm{d}x\, f(x)$ and $\displaystyle\sum_{n=1}^{\infty} f(n)$ where $f(x) \geq 0$ and decreases monotonically. Then, the series converges iff the integral converges.

4.6.2 Theorem

The sequence $a_k = \displaystyle\sum_{n=1}^{k} f(n) - \int_1^k \mathrm{d}x\, f(x)$ is convergent as $k \to \infty$ even if both the series and the integral diverge. Still, $f(x)$ here is assumed to be ≥ 0 and decrease monotonically.

Proof. Assume a positive and monotonically decreasing $f(x)$. So,

$$f(n) = \int_n^{n+1} \mathrm{d}x\, f(x).$$

So,

$$a_k = \int_1^{k+1} \mathrm{d}x\, f(x) - \int_1^{k} \mathrm{d}x\, f(x) = \int_k^{k+1} \mathrm{d}x\, f(x) \geq 0.$$

As we know that $f(x) \geq 0$, $a_k \geq 0$ (bounded from below). Also,

$$a_{k+1} - a_k = f(k+1) - \int_k^{k+1} \mathrm{d}x\, f(x) \leq 0.$$

Therefore, the sequence is monotonically decreasing $\Rightarrow \{a_k\}$ converges. $\square$

4.7 Summary

$$\text{Series} \begin{cases} \text{divergent} \\ \text{convergent} \;\Leftrightarrow\; \text{Cauchy series} \end{cases} \tag{52}$$

$$\begin{cases} \text{Convergence} & \left| s - \sum_{k=1}^{n} a_k \right| < \epsilon,\ \forall \epsilon > 0 \text{ if } n > N \\ \text{Cauchy series} & \left| \sum_{k=n}^{m} a_k \right| < \epsilon,\ \forall \epsilon > 0 \text{ if } n, m > N \end{cases} \tag{53}$$

$$\text{Tests of convergence} \begin{cases} \text{Comparison:} & \sum a_k \text{ converges if } \sum c_k \text{ converges } (c_k > |a_k|) \\ \text{Ratio test:} & \rho = \lim_{n\to\infty} \left| \frac{a_{n+1}}{a_n} \right| \\ \text{Root test:} & \rho = \lim_{n\to\infty} |a_n|^{1/n} \\ \text{Alternating series:} & a_n \text{ is monotonically decreasing to } 0 \\ \text{Integral comparison:} & \sum f(n) \text{ and } \int_1^{\infty} \mathrm{d}x\, f(x) \\ & (f(x) \geq 0 \text{ and decreases monotonically}) \end{cases} \tag{54}$$

5 The Topology of the Reals

5.1 Open and Closed Intervals

An open interval of $\mathbb{R}$ is defined as

$$(a, b) = \left\{x \middle| x \in \mathbb{R},\ a < x < b\right\}. \tag{55}$$

Note that the entire real line, $\mathbb{R}$, $(-\infty, \infty)$ is topologically an open interval.

A closed interval of $\mathbb{R}$ is defined as

$$[a, b] = \left\{x \middle| x \in \mathbb{R},\ a \leq x \leq b\right\}. \tag{56}$$

5.2 Closed Sets

Consider a set $S \subset \mathbb{R}$. A point $\bar{x}$ is called a **limit point** of S if there exists a sequence $x_n \in S$ which converges to $\bar{x}$.

A set S is **closed** iff it contains all its limit points. Any sequence in S cannot converge to a point outside S.

5.3 Open Sets

- A point $x \in \mathbb{R}$ is an **interior point** of a set $S \subset \mathbb{R}$ if there exists a neighborhood of x in S. Alternatively,

- A set S is said to be **open** if all its points are interior points:

$$\forall x \in S,\ \exists \epsilon > 0 \text{ such that } (x - \epsilon, x + \epsilon) \subset S. \tag{57}$$

  The set A is open if its complement A' is closed (and vice versa). As an example, $\mathbb{R}$ is both open and closed, while $\mathbb{Q}$ is neither.

- The closure of a set $S \subset \mathbb{R}$ is defined to be the union of S and all its limit points, denoted as $\bar{S}$. Clearly, $\bar{S}$ is closed.

- If $A \subseteq B$, then we say A is dense in B if $\forall x \in B$, $\exists y \in A$ such that $|x - y| < \epsilon$, $\forall \epsilon > 0$.

  As an example, $\mathbb{Q}$ is dense in $\mathbb{R}$.

- Equivalently, A is dense in B if B is the closure of A ($\bar{A} = B$).

5.4 Unions and Intersections of Open and Closed Sets

Let

$$A = \bigcup_{k=1}^{n} S_k, \qquad B = \bigcap_{k=1}^{n} S_k. \tag{58}$$

Unions of closed sets are related to intersections of open sets:

$$A' = \bigcap_{k=1}^{n} S_k'. \tag{59}$$

- The union of an infinite number of open sets is open;
- The intersection of an infinite number of closed sets is closed;
- The intersection of an infinite number of open sets is not necessarily open;
- The union of an infinite number of closed sets is not necessarily closed.

For instance,

$$I_n = \left[\frac{1}{n}, 1\right], \qquad A = \bigcup_{n=1}^{\infty} I_n = (0, 1]\,;$$
$$J_n = \left(-\frac{1}{n}, 1 + \frac{1}{n}\right), \qquad B = \bigcap_{n=1}^{\infty} J_n = [0, 1]\,.$$

6 Continuous Functions of Real Variables

6.1 Continuity

6.1.1 Definition: Continuity at a Point

By "continuity at point a", we mean

$$\forall \epsilon > 0,\ \exists \delta > 0 \text{ such that } |f(x) - f(a)| < \epsilon \text{ if } |x - a| < \delta. \tag{60}$$

Typically, given a specific function, we would end up finding $\delta(\epsilon)$.

A function f is said to be continuous on the interval I if it is continuous at every point $x \in I$.

6.1.2 Theorem

Suppose $f : \mathbb{R} \to \mathbb{R}$ is continuous at a. Then,

- $\exists \eta > 0$ such that f is bounded in $(a - \eta, a + \eta)$.
- If $f(a) > 0$, then $f(x) > 0$ in a neighborhood of a.

Proof. By definition, $\forall \epsilon > 0$, $\exists \delta > 0$ such that $|f(x) - f(a)| < \epsilon$ if $|x - a| < \delta$. This means that

$$a - \delta < x < a + \delta \quad \Rightarrow \quad f(a) - \epsilon < f(x) < f(a) + \epsilon.$$

- By choosing $\epsilon = 1$, clearly $f(x)$ has both an upper and a lower bound in the open interval $(a - \delta, a + \delta)$.
- Choose ϵ to keep $f(x)$ sufficiently close to $f(a)$ so that $f(x)$ is positive.

  As an example, $\epsilon = f(a)/2$ does the job: $f(x) > f(a)/2 > 0$ in the interval $(a - \delta, a + \delta)$.

□

6.2 Continuity and Sequences

Theorem: (proof: choose ϵ wisely) Consider $f : I \to \mathbb{R}$. Then, (A) and (B) are equivalent.

- (A) f is continuous at $a \in I$.
- (B) For every sequence a_n in I converging to a, the sequence $f(a_n)$ converges to $f(a)$.

Proof. In order to prove equivalence, we want to see

$$\text{(A) true} \ \Rightarrow\ \text{(B) true, and (A) false} \ \Rightarrow\ \text{(B) false.} \tag{61}$$

- (A) true $\Rightarrow$ (B) true:

  We are given the definition of continuity: $\forall \epsilon > 0$, $\exists \delta > 0$ such that $|f(x) - f(a)| < \epsilon$ if $|x - a| < \delta$.

  Set $x = a_n$, where $a_n \to a$. So, $\forall \delta > 0$, $\exists N(\delta)$ such that $|a_n - a| < \delta$ if $n > N$.

  Then, $n > N$ implies $|f(a_n) - f(a)| < \epsilon$. So, $f(a_n)$ converges to $f(a)$.

- (A) false $\Rightarrow$ (B) false: (A) is false implies that $\exists \epsilon > 0$ such that $|f(x) - f(a)| \geq \epsilon$ by taking x sufficiently close to a.

  Take $x = a_n$ where $a_n \to a$. So, a_n may be arbitrarily close to a yet $|f(a_n) - f(a)| \geq \epsilon$. So, $f(a_n)$ does not converge to $f(a)$.

The above proves that (A) and (B) are equivalent. $\square$

6.3 Fixed Point Theorem

6.3.1 Theorem: Bolzano

Let $f : [a, b] \to \mathbb{R}$ be a continuous function such that $f(a) < 0$ and $f(b) > 0$. Then $\exists c \in [a, b]$ such that $f(c) = 0$.

6.3.2 Theorem

Let $f : [a, b] \to [a, b]$ be a continuous function. Then $\exists \xi \in [a, b]$ such that $f(\xi) = \xi$.

6.4 Differentiability

6.4.1 Derivative

By "differentiable at a", we mean

$$\forall \epsilon > 0,\ \exists \delta > 0 \text{ such that } \left| \frac{f(x) - f(a)}{x - a} - f'(a) \right| < \epsilon \text{ if } |x - a| < \delta. \tag{62}$$

The derivative at a point a is defined as

$$f'(a) = \lim_{x \to a} \frac{f(x) - f(a)}{x - a}. \tag{63}$$

6.4.2 Theorem

Differentiable at $a \Rightarrow$ continuity at a.

Proof. We may write

$$f(x) - f(a) = \frac{f(x) - f(a)}{x - a}(x - a),$$

where the fraction is the derivative without a limit. As we know f is differentiable at a, we can take the limit $x \to a$ throughout. So,

$$\lim_{x \to a} f(x) - f(a) = \lim_{x \to a} f'(a) \cdot (x - a) = 0, \quad \lim_{x \to a} f(x) = f(a).$$

This last equality proves that $f(x)$ is continuous at a. $\square$

6.4.3 Higher Derivatives

$$f''(a) = \lim_{x \to a} \frac{f'(x) - f'(a)}{x - a} \tag{64}$$

6.4.4 Definition

Let $A \subseteq \mathbb{R}$, $f : A \to \mathbb{R}$, and $n \in \mathbb{N}$.

- f is C^0 if it is continuous on A.
- f is C^n differentiable if all its $f^{(k)}(x)$ derivatives are continuous on A for integers $0 < k \leq n$.
- f is C^∞ or smooth if $f^{(n)}(x)$ are continuous for all $n \in \mathbb{N}$.

6.5 Important Theorems

6.5.1 Extreme Value Theorem

If f is continuous on the interval $[a, b]$, then $\exists x_1,\, x_2 \in [a, b]$ such that f is max at x_1 and min at x_2.

6.5.2 Mean Value Theorem: MVT

If f is continuous on $[a, b]$ and differentiable on (a, b), then $\exists \xi \in (a, b)$ such that

$$\frac{f(b) - f(a)}{b - a} = f'(\xi) \quad \Rightarrow \quad f(b) = f(a) + (b - a) f'(\xi). \tag{65}$$

Proof. Define $h(x) = f(x) - f(a) = (x - a)K$, where $f(b) = f(a) + (b - a)K$. Clearly, $h(a) = 0 = h(b)$.

This implies that either $h(x) = 0$ for all x, or $\exists \xi \in [a, b]$ such that $h'(\xi) = 0$.

Thus, $h'(x) = f'(x) - K$, $h'(\xi) = f'(\xi) - K = 0$, $K = f'(\xi)$. Then, $h(b)$ gives the result as required:

$$f(b) - f(a) - (b - a) f'(\xi) = 0.$$

$\square$

6.5.3 2nd Order MVT

If f is continuous on $[a, b]$ and twice differentiable on (a, b), then $\exists \xi \in (a, b)$ such that

$$f(b) = f(a) + (b-a)f'(a) + \frac{1}{2}(b-a)^2 f''(\xi). \tag{66}$$

6.6 Taylor Series and Analytic Functions

6.6.1 Taylor's Theorem: nth Order MVT

$$f(b) = \sum_{k=0}^{n-1} \frac{(b-a)^k}{k!} f^{(k)}(a) + \frac{(b-a)^n}{n!} f^{(n)}(\xi) \tag{67}$$

6.6.2 Taylor Series

Suppose f is C^∞ on interval A. The Taylor series about a point $a \in A$ is:

$$f(x) = \sum_{k=0}^{\infty} \frac{(x-a)^k}{k!} f^{(k)}(a). \tag{68}$$

6.6.3 Definition

Suppose f is C^∞ on interval I. Then f is **real analytic** at a point $a \in I$ iff f is equal to its Taylor expansion in some open interval containing a:

$$\exists \epsilon > 0 \text{ such that } f = f_{\mathrm{T}} \text{ in } (a-\epsilon, a+\epsilon). \tag{69}$$